

ALPACA AI TRADING AGENTS HACKATHON · LABLAB.AI · 28 AUG – 4 SEP 2026

# IV DESK

An autonomous options desk that sells overpriced volatility — and stands down the rest of the time.



PAPER TRADING ONLY

ACCOUNT PA39HSCQE8S3

[GITHUB.COM/ALVAAROCL/IV-DESK](https://github.com/ALVAAROCL/IV-DESK)

## THE IDEA

# An insurance company for the stock market

- Defined-risk structures only — iron condors, 1–3 days to expiration
- Sells only when implied vol sits above a realized-vol forecast (VRP)
- Sells only when dealer gamma favors range-bound behaviour

## WHAT MAKES THIS DIFFERENT

Most agents predict price. This one reads the option surface — and the product isn't the trades. It's the refusals, each written to the journal with the number that caused it.

295 stand-downs • 1 trade — both the strategy working

# The LLM never touches the money path

| LAYER              | DETERMINISTIC? | WHAT IT DOES                                                                                   |
|--------------------|----------------|------------------------------------------------------------------------------------------------|
| Signal             | yes            | VRP ratio (IV vs. RV forecast), normalized dealer gamma, regime, skew                          |
| Structure & sizing | yes            | Strike selection at a target delta behind liquidity gates; contracts capped to the risk budget |
| Risk Officer       | yes            | Every gate, no discretion — evaluate() returns (ok, reason), no LLM path in                    |
| The desk (LLM)     | bounded        | Four seats debate whether to open a trade already approved — trim, veto, or nothing            |
| Execution & exits  | yes            | 4-leg mleg order via the Alpaca CLI; 50% TP / 2× stop / expiry close                           |
| Journal            | yes            | Append-only JSONL — every signal, refusal, debate transcript, open and exit                    |

THE LINE THAT MAKES IT STRUCTURALLY SAFE — NOT A PROMPT PROMISE

```
contracts = max(0, min(int(head.contracts), cap))    # agent/debate.py
```



# Three gates. Any one stands the desk down.

1

## Volatility risk premium

ATM implied vol over a Yang-Zhang + EWMA realized-vol forecast, expressed as a ratio — holds across any vol regime.

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Gate: `ratio ≥ 1.05`

2

## Dealer gamma (GEX)

Normalized from chain open interest — a regime signal, never a precision input — with a dead zone near zero.

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Positive gamma dampens moves

3

## Regime

ADX / EMA read. A trending tape stands the desk down rather than fading it.

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Short premium into a trend is how books die

# Four seats, all open models, on Featherless

## Quant ensemble

Up to 3 open models vote independently; needs a strict majority of the models dispatched, not of those that answered.

## Bull / Bear

Must anchor claims to real numeric signal fields, named; an argument citing fewer than two is discarded outright.

## Desk Head

Sizes the trade ( $\leq$  the Risk Officer's cap) and writes a falsifiable prediction: a closing range, on a date.

NO PATH WHERE GARBAGE MEANS YES

Outage, timeout, truncated JSON, a split ensemble — every failure mode resolves to approved=False, reason recorded.

90s debate budget · DESK\_DEBATE=off is the kill switch · default is required

DISCIPLINE

174 opportunities. 11 trades.



+\$484

HELD-TO-EXPIRY P&L

8 / 11

WINNERS (BACKTEST)

78 → +\$401

EVERY GATE DISABLED – EDGE REAL, BUT THIN

No honest lever buys more trades without hurting the expectancy — this is a strategy whose discipline is auditable.



# Two real vetoes this week

2 SEP

## SPY cleared VRP, gamma vetoed it

Ratio 1.61 (well above the 1.05 floor) — but GEX read  $-0.106$  (dealers short gamma), so the signal layer stood it down before the desk was due to see it. It ran anyway in a shadow debate: 2 of 3 Quant models confirmed, Bull and Bear argued genuinely opposite cases, and the Desk Head overruled its own Quant majority.

*"The GEX signal must be strong enough to overcome the high ATM IV and choppy regime for this trade to win."*

3 SEP

## Every gate cleared, the credit floor didn't

VRP 1.74, GEX  $+0.58$  (long gamma — exactly the regime the strategy sells into), sized to 7 contracts by the Risk Officer — and still rejected before reaching the desk: credit/width came back at 0.10 against the 0.20 floor.

*A ODTE condor simply doesn't carry the credit that floor was calibrated on.*

RESULTS · MON 31 AUG → THU 3 SEP

1 trade, 1 win, +\$318.85

ENDING EQUITY

\$100,318.85

+\$318.85 · +0.32% · four sessions

QQQ iron condor

8 contracts · strikes 713/715P–720/722C · \$352 credit · \$1,248 max loss. Opened Thu 3 Sep 11:58 ET after clearing every gate and the real four-seat debate — the mesa's only live approval of the week. Closed inside every strike.

295

DOCUMENTED STAND-DOWNS

360

SIGNAL EVALUATIONS THIS WEEK

4

DEBATES RUN, 1 APPROVAL

1 / 1

PREDICTIONS CORRECT

Desk Head's prediction: QQQ closes in [705, 730] on 2026-09-03 — correct.



# 181 tests — and the bug we didn't hide

- Four-seat debate tested end-to-end against injected transport doubles — no network, no API keys
- Black-Scholes IV/greeks backfill (23 tests) — built mid-week when Alpaca turned out not to publish greeks for same-day options
- Strike selection, exit manager, risk gates and the account guard all tested against synthetic data

## Issue #26 — the real incident

The leg-closing routine flipped a leg's side but not its matching position-intent prefix. The one real trade this week hit it live — 13 failed close attempts, 13:28–15:13 ET — before the position expired inside the strikes and OCC settled it automatically.

Root-caused from the live error log, fixed, and pinned with 4 regression tests the same afternoon. Left in the record — a desk that documents its own bugs the same way it documents its stand-downs is the point of this project.

# Deep, real use of the platform

## Trading API through the official CLI

every trading call in the production loop shells out to `alpaca api METHOD /path`, pinned and version-verified in CI

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## order\_class: mleg

a 4-leg iron condor as one order, signed `limit_price` (negative = credit), zero-price rejected outright

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## Options market data over REST

chain snapshots with greeks and IV, per-contract open interest, SIP daily bars

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## Idempotent and self-healing

deterministic `client_order_id` per trade intent; every tick reconciles against Alpaca as the source of truth

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## Paper only

competition account PA39HSCQE8S3, \$100,000, first order Mon 31 Aug 09:30 ET, untouched before that

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**The journal is in the repo — every signal, every refusal, every trade.**

`github.com/alvaarocl/iv-desk`

`alvaarocl.github.io/iv-desk`

Account PA39HSCQE8S3